

The Cycles agroecosystem model: Fundamentals, testing, and applications

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ABSTRACT

Models of the soil–plant–atmosphere continuum are repositories of knowledge and gears in analytical and decision support tools applied to agroecosystems. In this paper, we present the Cycles agroecosystem model theory along with test cases and applications. Cycles combines innovations for simulating soil hydrology and biogeochemistry, including carbon and nitrogen saturation theory, with a modular software architecture. These elements enable simulating monoculture or polyculture crop sequences and associated management practices, containerizing for applications that require high-performance computing, and data assimilation at runtime. A comparison of simulated and measured daily evapotranspiration (ET) obtained with the eddy covariance method for maize (*Zea mays* L.) and shrub willow (*Salix* spp.) shows that Cycles represents well meteorological and vegetation controls of ET (root mean square error or RMSE = 0.75 mm d⁻¹). Cycles accurately simulated differences of 150 mm in growing season ET between these two plant communities. Comparisons of modeled versus measured soil water content under soybean (*Glycine max* [L.] Merr.) in southeastern Pennsylvania for six soil layers at 0.1-m increments show accurate representation of water depletion and recharge (RMSE of 0.027–0.011 m³ m⁻³). Simulations of growth and nitrogen uptake of wheat (*Triticum aestivum* L.) in eastern Washington also highlight the model's skill simulating processes that affect water and nutrient fluxes simultaneously. To highlight Cycles' suitability for incorporation in high performance computing applications, we present a coupling of Cycles with an autonomous crop sequence builder (Cycles-A) in the Chesapeake Bay watershed. This system automatically identified areas for double cropping and selected the optimum combination of annual crops across the watershed. The Cycles model innovations and agroecosystem framing continue advancing the premise of making models not only dynamic knowledge repositories but useful tools for research and landscape management.

1. Introduction

Models of the soil–plant–atmosphere continuum (SPAC) have become both a repository for our understanding of functional soil physics, biogeochemistry and plant community ecophysiology, and the kernel of decision support tools in agriculture (Zhai et al., 2020). One category of these models, that of cropping system models that simulate a notional 1D field, is well represented by EPIC (Williams, 1995), ECOSYS (Grant, 1997), DSSAT (Jones et al., 2003), APSIM (Keating et al., 2003), and CropSyst (Stöckle et al., 2014), among other models.

These models are built with a comparable general structure. The 1D

field is constrained by boundary conditions at the bottom of the soil profile and at a reference height in the atmosphere. Soil and air are discretized in layers, each with compartments for scalars that can flow between layers. Vegetation straddles and connects these layers and can transfer energy and mass between them. Operations like irrigation, fertilization or tillage alter fluxes and system properties. Beyond this generic structure, there are differences in how models represent soil and plant processes. These differences can be acute. As a result, cropping system models running on the same inputs produce divergent outcomes (Bassu et al., 2014; Camargo and Kemanian, 2016; Kimball et al., 2019). To be useful beyond trivial applications, agroecosystem models must

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represent with increasing fidelity the interactions between soil, plant, climate, and other biotic and abiotic factors that control agroecosystem performance and consider, as well, users' demands in their design (Antle et al., 2017; Zhai et al., 2020).

Current cropping system models evolved either as bottom-up or top-down models that steadily expanded their domain of interest. Conceptually, bottom-up models build towards aggregated outputs that result from simulations at lower hierarchical levels, while top-down models let the desired output direct the underlying model structure, lumping processes into empiric parameters. For example, carbon dioxide (CO₂) accrual by vegetation can be calculated starting from chloroplast-level photosynthesis (bottom up, e.g., Collatz et al., 1991) or by using a big-leaf to simulate radiation absorbed and a multiplier that converts radiation to biomass (top down, e.g., Warren Wilson, 1967; Monteith, 1977). No SPAC or cropping system model is strictly bottom-up or top-down. In addition, and paraphrasing Thornley and France (2007), a teleonomic slant is often apparent when modelers replicate observed behaviors rather than let them emerge from fundamentals. A conundrum in the life sciences is that bottom-up models are mechanistically attractive but cannot be easily generalized, while generic top-down models cannot handle all the specifics. As model applications expand and priorities shift, maintaining a competent balance between these approaches is challenging.

The bottom-up approach is exemplified by ECOSYS (Grant, 1997). In contrast, EPIC (Williams, 1995) follows top-down and teleonomic approaches. The EPIC development included valuable innovations, such as the representation of plant species and cultivars through generic parameters as advocated during the first decades of crop model development (Acock and Reynolds, 1988). The DSSAT system (Jones et al., 2003) has the flavor of crop-specific models emphasizing ecophysiology and blending bottom-up and top-down approaches, as reflected for example in its components CERES-MAIZE (Jones and Kiniry, 1986), the generic CropGro for grain legumes (Boote et al., 2018), and OILCROP-SUN (Villalobos et al., 1996). APSIM followed a similar trajectory (Keating et al., 2003), with applications emphasizing plant ecophysiology and phenotypic prediction (e.g., Messina et al., 2018). CropSyst's (Stöckle et al., 1994; Stöckle et al., 2003, 2014) foundation is the SPAC biophysics from Campbell (1985). There are other analogous modeling systems and growing applications through global models (Drewniak et al., 2013). However, these modeling systems retain software design features and disciplinary emphases that constrain their application domain.

As this type of models evolve, modeling teams both converge and diverge in their approaches to represent biophysical processes. Convergence can reflect collective efforts to fill knowledge gaps (e.g., Maiorano et al., 2017) or, alternatively, a transient consensus rather than an agreement on fundamentals. An example of the latter is the adoption by multiple teams of the Century model (Parton et al., 1994) to represent soil C cycling (Gijssman et al., 2002; Izaurrealde et al., 2006; Oleson et al., 2013; Stöckle et al., 2013; Zhang et al., 2013). While incorporating Century, these teams overlooked its simplifications by design, such as the lumping of processes in a notional 0.23-m soil layer with no stratification of soil attributes. In addition, the conceptual framework of Century, traceable to the Phoenix model (McGill et al., 1981), was not an end point, but one that was being reviewed in light of advances in soil biogeochemistry (Šmejkalová and Piccolo, 2008; Stöckle et al., 2012; Sutton and Spósito, 2005; Hedges et al., 2000; Burdon, 2001; Wrobel et al., 2003; Schmidt et al., 2011). Divergence among modeling approaches can reflect genuine differences in the conceptualization of biophysical processes. Camargo and Kemanian (2016) compared algorithms used to simulate plant water uptake in six modeling systems and found that differences among them are relevant but difficult to justify, because the fundamentals of water flow in SPAC have been unchallenged for decades. Lack of agreement in simulating a fundamental process like evapotranspiration (ET, Kimball et al., 2019), also illustrates this point.

Some processes are not incorporated in models due to incomplete knowledge of the subject matter. For example, cropping system models lack algorithms to represent the interactions lumped in concepts such as the "rotational effect" (Russell, 1987; Ernst et al., 2018). Examples of rotational effects are the suppression of cryptic soil diseases through crop rotation (Angus and van Herwaarden, 2001), enhancement of cereal crops yield when alternating with legumes in a rotation (Seifert et al., 2017), and changes in soil physical properties due to specific crop sequences or soil management (Ernst et al., 2016, 2018; Stefani-Faé et al., 2020). Cropping system models need to improve, as well, the representation of soil-phytobiome-plant feedbacks (Kakouridis et al., 2022) and 3D landscape level processes (Stöckle and Kemanian, 2020). And from an application perspective, these models need to conform to modern standards in which they are but one piece in complex data-model workflows (Leonard and Duffy, 2016; Zhai et al., 2020; Gil et al., 2021).

In summary, existing cropping systems models differ, most systems have limited skill at representing agroecosystem level responses beyond the agronomic performance of certain plant species, and the design of current models might not facilitate incorporation in data-model workflows. Among the motivations to develop Cycles is to offer a platform that facilitates overcoming these limitations by incorporating or assembling in one model both new knowledge and functional algorithms in a code structure that can be continually improved and seamlessly incorporated in data-model workflows.

Our goal in this paper is to present the most salient features of the Cycles agroecosystem model along with model testing and applications. We use the term agroecosystem to highlight the importance of a broad set of ecological and management interactions at field and landscape scales. Cycles combines a contemporary understanding of soil hydrology, biogeochemistry, and plant growth, and an intuitive agronomic input structure that facilitates simulating any monoculture or polyculture sequence in combination with multiple management practices. Cycles' object-oriented structure is suitable not only for agroecosystem analysis but for expansion into landscape-level modeling and for containerizing. The paper is divided into three parts. The Methods section presents the theoretical underpinning that distinguishes Cycles from existing modeling systems. The Results section presents evaluations of the model against field observations of variables of importance and applications to illustrate the model's capabilities. The Discussion highlights strengths, limitations, and future developments. The [supplemental information \(SI\)](#) document provides further background on the model and on the data used in the examples and a link to the freely available manual, executable, and simulation examples in GitHub (SI Section I).

2. Methods

2.1. Cycles model general structure

Cycles is a biophysically-based, multi-year, and multi-species agroecosystem modeling system. Its design followed these criteria: (i) balance bottom-up and top-down approaches to retain mechanisms and favor functionality, transparency, and a minimum of calibration parameters; (ii) favor a generalized representation of elements and fluxes in SPAC over taxonomic soil- and plant-specific nuances; (iii) represent management interventions like tillage, fertilization, and irrigation through generic inputs and algorithms; (iv) use a modular structure, so its components or the whole can be treated as objects that can be instantiated to enable further development of physically distributed models (coupling groundwater, water reservoirs, and stream networks); and (v) enable data assimilation at run-time, previewing applications in complex data-model workflows.

The model is conceived as the coupling of the energy and water balance with the cycles of carbon (C), and nitrogen (N) (Figs. 1 and 2). Fluxes into and out of a pool define each differential equation. Intensive

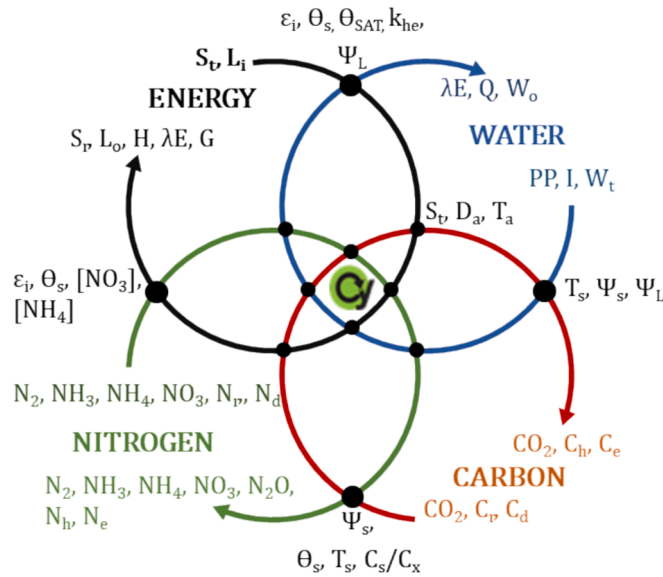


Fig. 1. Schematics of the coupling of the energy, water, C, and N cycles, with fluxes in (arrow starts) and out (arrow ends) for each component. Internal fluxes not represented. The nodes at the arrows' intersections represent transfers of information. Fertilization, tillage, and irrigation alter inputs and intensive properties; harvest alters output rates. Energy enters as shortwave (S_t) and thermal radiation (L_i), and is dissipated as reflected S_r (S_r), thermal outgoing radiation (L_o), sensible heat (H), latent heat (λE), or stored (G). Shaded text indicates fluxes that are currently not simulated. Liquid water enters through precipitation (PP), irrigation (I), and capillary rise (W_t), and leaves via λE , runoff (Q), and soil percolation or lateral flow (W_o). Carbon enters through photosynthesis, soil amendments (C_e), and erosion deposition (C_d), and leaves via respiration, harvest (C_h), and erosion (C_e). Other symbols in Table 1. Details on the C and N cycles in Fig. 2.

properties such as temperature (T), water potential (ψ), and N concentration in soil or plant tissue regulate fluxes. These fluxes are resolved with a forward Eulerian integration constrained by mass conservation. A list of relevant state variables is provided in Table 1.

Cycles shares biophysical fundamentals with CropSyst (Stöckle et al., 2014) that are derived from Campbell (1985), incorporates innovations in biogeochemistry such as the concept of C and N saturation (Kemanian et al., 2005; Kemanian and Stöckle, 2010; White et al., 2014) and plant growth based on stomata optimization theory (Cowan, 1977; Katul et al., 2010), and represents plant species in annual and perennial polycultures explicitly (Pravia et al., 2019).

2.2. Energy balance

The energy fluxes are solved in 1D and at a daily time step. Like most models, a daily weather file provides forcing weather conditions. The weather variables are precipitation (PP , $\text{kg m}^{-2} \text{d}^{-1}$), maximum and minimum temperature (T_x and T_n , $^{\circ}\text{C}$), solar radiation (S_t , $\text{MJ m}^{-2} \text{d}^{-1}$), maximum and minimum relative humidity (RH_x and RH_n , %), and wind speed (WS , m/s). The radiative energy input is from solar radiation (input data) and atmospheric long wave radiation (estimated from air temperature and humidity as in Allen et al., 1998). During the simulation, surface conditions do not alter air temperature, humidity or windspeed. Energy advection is implicit in the weather record.

The energy balance terms that are solved explicitly are the latent heat flux and the soil heat storage. These are, however, not solved simultaneously or using an equilibrium surface temperature. The Penman-Monteith equation, which implicitly includes the net radiation, is used to estimate the evaporative demand. This demand is allocated to the canopy, residues, snow, and the soil based on their corresponding radiation interception. This is a shortcut that bypasses the calculation of each surface temperature as done in the SHAW model (Flerchinger, 2000) and the CUPID model (Norman et al., 1990) but renders a

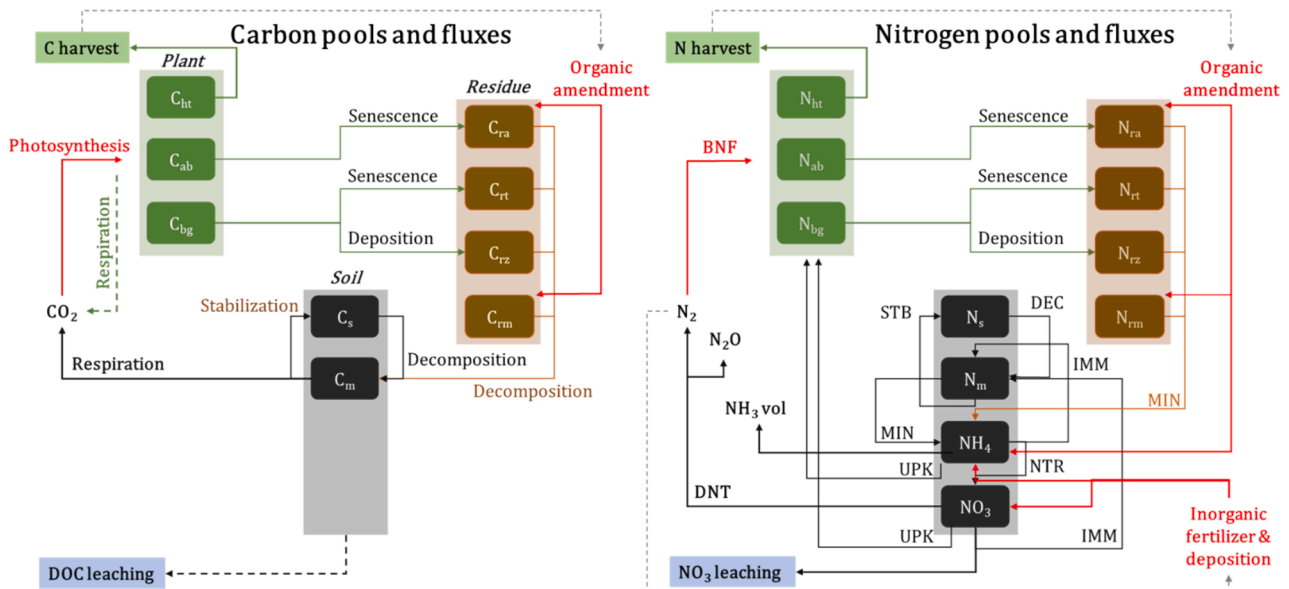


Fig. 2. Diagram of carbon (C) and nitrogen (N) pools and fluxes in plant, residue, and soil. Solid lines and arrows represent fluxes and flux direction, respectively. Green represents vegetation pools, brown soil residue pools, and black soil organic C and microbial biomass C. Dashed lines are fluxes that are not modeled but conceptually close the C and N cycle, making inputs to and outputs from the system clear. The C cycle starts with $\text{CO}_2\text{-C}$ entering the plant pool through photosynthesis or C entering the soil residue pool through the addition of organic amendments. Autotrophic and heterotrophic respiration return $\text{CO}_2\text{-C}$ to the atmosphere. Analogously, mineral N enters the system through biological N fixation, fertilization, or deposition, and leaves the system through harvest, leaching, volatilization of $\text{NH}_3\text{-N}$ and denitrification of $\text{NO}_3\text{-N}$. Losses of N_2O through nitrification are modeled but not represented in the diagram for clarity. User-defined operations and intensive properties like temperature and soil and plant water potential control fluxes. Abbreviations: BNF = biological N fixation; DEC = microbial decomposition; DNT = denitrification, IMM = microbial uptake of mineral N, MIN = mineralization of organic N, NTR = nitrification of NH_4 , and UPK = plant uptake of mineral N.

Table 1

Partial list of Cycles' pools, reference rates and other parameters, and variables related to carbon (C) and nitrogen (N) cycling. All organic C pools have a corresponding organic N pool. Soil pools are represented in each soil layer.

Name	Unit	Symbol	Notes; dependences
Plant pools			
Shoot C, N	kg m ⁻²	C _{ab} , N _{ab}	Leaf, stem, unharvested
Root C, N	kg m ⁻²	C _{bg} , N _{bg}	Excludes rhizodeposition
Grain / underground C, N	kg m ⁻²	C _{ht} , N _{ht}	Grain or underground harvestable tissue
Plant parameters			
Base temperature for development	°C	T _b	Cultivar dependent; acclimation
Thermal time to flowering	°C d	TT _{flo}	Cultivar / photoperiod / vernalization
Thermal time to maturity	°C d	TT _{mat}	Cultivar dependent
Crop coefficient ^(a)	Unitless	k _c	Lumps surface conductance and roughness
Maximum transpiration	kg m ⁻² d ⁻¹	TR _x	Maximum plant liquid water flow
Onset of stomatal closure leaf ψ ^(b)	J kg ⁻¹	ψ_{Lo}	
Stomata closure leaf ψ	J kg ⁻¹	ψ_{Lf}	
Maximum canopy cover	Unitless	ϵ_{lc}	< 1 in widely spaced rows
Radiation use efficiency	g MJ ⁻¹	ϵ_r	Biomass gain per unit of intercepted ϵ ^(c)
Transpiration use efficiency ^(d)	g kg ⁻¹	ϵ_w	Biomass gain per unit of water transpired
Soil pools			
Soil organic C, N	kg m ⁻²	C _s , N _s	Soil organic C, N
Microbial C, N	kg m ⁻²	C _m , N _m	Soil microbial C, N
Manure C, N	kg m ⁻²	C _{rm} , N _{rm}	Manure C, N
Plant shoot residue C, N	kg m ⁻²	C _{ra} , N _{ra}	Residue aboveground C, N
Plant root residue C, N	kg m ⁻²	C _{rt} , N _{rt}	Root residue C, N
Plant rhizodeposition C, N	kg m ⁻²	C _{rz} , N _{rz}	Root rhizodeposition C, N
Soil nitrate	kg m ⁻²	NO ₃ ⁺ - N	N mass (no oxygen included)
Soil ammonium	kg m ⁻²	NH ₄ ⁺ - N	N mass (no hydrogen included)
Soil volumetric water content	m ³ m ⁻³	θ_v	
Soil parameters			
Soil carbon saturation	g kg ⁻¹	C _{so}	C _s that saturates mineral soil; f(clay)
Reference C _s decay rate	kg m ⁻² d ⁻¹	k _s	C removed from soil organic matter pool
Reference C _m death rate	kg m ⁻² d ⁻¹	k _m	C removed from microbial biomass
Reference C _{ra} decay rate	kg m ⁻² d ⁻¹	k _{ra}	C removed from aboveground residues
Reference C _{rt} decay rate	kg m ⁻² d ⁻¹	k _{rt}	C removed from belowground residues
Reference C _{rz} decay rate	kg m ⁻² d ⁻¹	k _r	C removed from rhizodeposits
Reference C _{rm} decay rate	kg m ⁻² d ⁻¹	k _{rm}	C removed from aboveground manure
Maximum denitrification rate	kg kg ⁻¹ d ⁻¹	k _{nd}	N removed from NO ₃ pool; kg kg ⁻¹ soil
Maximum nitrification rate	kg kg ⁻¹ d ⁻¹	k _{an}	N removed from NH ₄ pool; kg kg ⁻¹ soil

^aMultiplier of ET_o to obtain ET_p .

^b ψ symbol for water potential.

^cTotal incoming daily solar radiation

^dAt $D = 1$ kPa and $P_{atm} = 101.325$ kPa.

^eReference rates at optimum temperature and moisture. They are not maximum rates, as soil disturbance by tillage can increase their magnitude above the reference.

reasonable estimate of evaporation from wet surfaces. To estimate the soil heat flux and storage, we estimate a surface temperature as a function of the prior day temperature, the air temperature, and snow and soil cover. The canopy does not store energy, and the net radiation that is not dissipated as latent heat or stored in the soil is therefore assumed to be sensible heat. In Cycles-L (Shi et al., 2023), the energy balance is closed in every sub-daily time step in large part to account for differences in topography and aspect which are critical to simulate soil temperature and snow cover and melt. Details are presented in SI Section II.

2.3. Water uptake

The maximum daily latent heat flux, expressed as vapor transfer, is estimated based on a reference crop evapotranspiration (ET_o , kg m⁻² d⁻¹), calculated with the Penman-Monteith equation as presented by Allen et al (1998). Adjustments depending on plant species or plant community properties are made by using the crop coefficient k_c , an empiric constant (range 0.8 to 1.2) that multiplies the ET_o and is weighted by the proportion of vegetation ground cover. The algorithm to simulate transpiration has been adapted from Campbell et al (1974), Campbell (1985), and Jara and Stockle (1998), and is described in Camargo and

Kemanian (2016). Water redistribution in the soil column due to hydraulic lifting (it can be downward) is an emergent property.

2.4. Soil water infiltration, runoff, redistribution, and percolation

Precipitation (liquid), irrigation, and snowmelt water are considered water inputs to the soil surface. This water is used first to wet soil residues. The remaining water is available for runoff and infiltration. Instead of directly tackling the solution of Richards equation with added macropore flow to solve for infiltration and water redistribution simultaneously, we implemented two alternative methods to select from that resolve infiltration or runoff first, and then deal with redistribution. To solve for infiltration, we implemented the Green and Amp method as described by La Follette et al. (2023) for layered soils. To solve for daily runoff first (Q , kg m⁻² d⁻¹) we implemented an adaptation of the Soil Conservation Service (now USDA-NRCS) Curve Number method (USDA, 1972). Details are presented in SI Section III. Any other approach will also contain empiricisms (see discussion).

Infiltrating water is redistributed to lower soil layers at a sub-daily time step based on the hydraulic conductivity and storage capacity of each soil layer. Hydraulic properties are determined by pedotransfer functions (Saxton and Rawls, 2006). First, water available for

infiltration is allocated to the first soil layer, up to saturation. Second, since the water potential gradient between adjacent soils layers with soil moisture above field capacity is mostly gravitational, the water flow between layers is assumed to depend on gravity g (g m s^{-2}) and an effective hydraulic conductivity (k_{he} , kg s m^{-3}), calculated as a capacitance-weighted k_h in the range between the current water content (θ_s) and field capacity:

$$k_{he} = \frac{\int_{\psi_{sfc}}^{\psi_s} k_s(\psi_e/\psi_s)^{2+3b} (-1/b\psi_s)\theta_s(\psi_e/\psi_s)^{1/b} d\psi}{\int_{\psi_{sfc}}^{\psi_s} (-1/b\psi_s)\theta_s(\psi_e/\psi_s)^{1/b} d\psi} \quad \text{for } \psi_s > \psi_{sfc} \quad (1)$$

The term inside the integral in the denominator is the capacitance $d\theta_s/d\psi_s$ with $\theta(\psi_s) = \theta_{sat}(\psi_e/\psi_s)^{1/b}$, where b is the exponent of the soil moisture release curve. The subscript *sat* stands for saturation. The term multiplying the capacitance in the numerator is the hydraulic conductivity and ψ_e is the air entry potential. The weighting by capacitance slows down water flow as the soil approaches field capacity. To accommodate differences in soil layer thickness, soil water content, and hydraulic properties, the time step is dynamically adjusted and set as that of the layer with the slowest travel time in the soil profile at the beginning of the time step. The travel time t (seconds) as a function of k_{he} and θ_s for each soil layer is:

$$t(k_{he}, \theta_s) = \frac{(\theta_s - \theta_{sfc}) \times dz \times \rho_w}{k_{he} \times g} \quad \text{for } \theta_s > \theta_{sfc} \quad (2)$$

Third, if the water allocated to daily infiltration has not been redistributed after 24 h, then the soil layers are saturated from top to bottom and excess water, if any, drained as percolation, conserving Q . The water flow in and out of each soil layer is stored and used to estimate solute transport.

Adjustable vertical and horizontal bypass coefficients can help account for macropore flow and tile drainage. The vertical bypass coefficient represents the fraction of the water volume above field capacity in a layer that can pass with no resistance to the layer immediately below, provided the receiving layer has room to accommodate the incoming water volume. It has limited effect on water flow compared to a lateral bypass of the same magnitude. The lateral bypass allows simulating tile drainage or lateral flow of water and solutes into an adjacent landscape position, leaving the soil profile as edge of field loss. Cycles-L (Shi et al., 2023), a landscape-level version of Cycles, requires no empiric bypass coefficient as macropore flow is parameterized and lateral flow resolved explicitly for each grid in a watershed.

2.5. Carbon balance – Plant CO₂ acquisition, biomass partitioning, and yield

Crop C acquisition by plant canopies is simulated using the resource capture and resource use efficiency approach. Plants capture radiation via the canopy and water mostly via the roots. These two captured resources are converted to potential growth by multiplying them by radiation and water use efficiency estimates, and the minimum of the two determines the growth rate. Thus, C acquisition is based on imposing both a limitation to photosynthesis by radiation capture and by CO₂ diffusion, the latter surrogated by water loss from leaves to air (i.e., transpiration). The approach is a simplified version of Farquhar et al (2001) photosynthesis model that conforms with stomata optimization theory (Cowan 1977; Hari et al., 1986; Lloyd 1991; Katul et al., 2010; Bassiouni and Vico, 2021), and thus departing slightly from that used in CropSyst (Stöckle et al., 2014).

A corollary of optimization theory is that the dependence of the leaf-level water-use efficiency (g biomass kg^{-1} water) on the leaf-to-air vapor pressure deficit (D_L) is proportional to $1/\sqrt{D_L}$. By extension, the canopy-level water-use efficiency is proportional to $1/\sqrt{D_a}$, where the prime denotes canopy. For practical purposes, we use the air vapor pressure

deficit, D_a , as surrogate for D_L , as air and leaf temperature are close except under high radiation load and low transpiration when D_a and D_L can differ by > 0.5 kPa (Taborski et al., 2022). In these conditions, limited growth prevents major errors in modeling carbon acquisition. The inverse square root dependence is close to the empiric formulation found by Kemanian et al (2005). Thus, under light limited conditions, the daily canopy growth rate (G_B , $\text{kg m}^{-2} \text{d}^{-1}$) is proportional to radiation intercepted by the canopy, and when radiation is no longer limiting, growth is proportional to transpiration. In other words, G_B is the minimum of growth limited by radiation (G_R) or growth limited by transpiration (G_T):

$$G_B = \min(G_R, G_T) \quad (3)$$

$$G_R(\epsilon_{ie}, S_t) = \epsilon_R \epsilon_{ie} S_t \quad (4)$$

$$G_T(D_a) = \frac{\epsilon_w}{\sqrt{D_a}} TR \quad (5)$$

The parameter ϵ_R is the radiation use efficiency (Warren Wilson, 1967; Monteith, 1977). A conversion allows using the more appropriate PAR based coefficient. Maximum ϵ_R for each species based on solar radiation are given in Table SI.2. Similarly, ϵ_w is the water use efficiency at $D_a = 1$ kPa and at $P_{atm} = 101.3$ kPa. Both ϵ_R and ϵ_w should be given for a reference atmospheric CO₂ concentration and scaled accordingly as CO₂ changes. For a discussion on the implications of optimization theory under changing CO₂ see Bassiouni and Vico (2021).

Partitioning of biomass to shoot (f_{st} , kg kg^{-1}) and roots is estimated allometrically (SI Section IV). Unlike models that further partition shoot biomass into leaf, stem, and other components, and require estimating the specific leaf area and senescence to calculate the delta leaf area index per unit of time, in Cycles we simulate the time evolution of canopy cover (ϵ_i) using a double sigmoid equation:

$$\epsilon_i(TT_f) = \frac{\epsilon_{ix}}{1 + e^{(a+b \times TT_f)} + e^{(c+d \times TT_f)}} \quad (6)$$

The ϵ_{ix} is a crop input parameter that equals 1 in most cases (100 % achievable canopy cover) but can be reduced to represent widely spaced rows or sparse canopies that never reach full canopy cover. The ratio $-a/b$ determines the TT_f at which ϵ_i is half of the maximum radiation interception (ϵ_{ix} , an input parameter) with a positive slope, and the ratio $-c/d$ the TT_f at which that threshold is reached with a negative slope during senescence (b and c are negative). The magnitude of a , b , c , and d affects the slope; default values of 6, -20 , -15 , and 16, respectively, are hard-coded and represent a normalized plant density (PD_f) of 1. A comparable approach with a different formulation is used by Steduto et al. (2009). The first derivative of Eq. (6) is used to estimate the rate of change of ϵ_i as a function of TT_f . The rate is reduced by N stress. Haying, grazing, tillage, and cold damage reduce ϵ_i and TT_f , rejuvenating the canopy. That makes simulating haying and grazing simple and generalizable.

To represent faster canopy cover progression due to increased plant density or due to species or cultivars intrinsic behavior (e.g., canola, *Brassica napus* L.), the plant density can be adjusted upward. A higher plant density (for example $PD_f = 1.2$) hastens the onset of the exponential increase. This is accomplished by adjusting ϵ_i (Eq. (6)) to an effective one (ϵ_{ie}) for a given species in the community:

$$\epsilon_{ie}(PD_f) = 1 - e^{(\ln(1-\epsilon_i)\sqrt{PD_f})} \quad (7)$$

When dealing with polycultures, the density of each species can be adjusted proportionally (for example, $PD_f = 0.33$ for three species, so that the sum approaches 1), or overseeded using higher densities. This can be useful when dealing with polycultures of perennial pastures.

Grain yield (or that of subterranean structures) is calculated based on a dynamic estimation of the harvest index (HI , ratio of grain to aboveground biomass at maturity, kg kg^{-1}) based on Kemanian et al.

(2007). Crop N demand is estimated using the N dilution approach (Lemaire et al 1990; Stöckle and Debaeke, 1997; Gastal and Lemaire 2009). The N concentration in the biomass of new growth is calculated based on the N concentration for a crop growing without water stress, which indirectly adjusts N demand for phenology, and is also adjusted by plant density. Luxury N uptake is allowed, but there is no compensatory N uptake to satisfy past N demand. Biological N fixation by legumes assumes that N demand above that supplied by the soil is met through N fixation without luxury consumption. The partitioning of N to harvestable structures follows Kemanian et al. (2007).

2.6. Soil carbon and nitrogen balance – Saturation theory

The soil has the following six organic C and N compartments, excluding living plants: microbial pool, soil organic matter pool, and residue pools representing dead root mass, dead shoot mass, rhizodeposits, and manure (Fig. 2; Table 1). Soil C and N cycling are based on C saturation theory (Hassink and Whitmore, 1997; White et al., 2014), which states that there is an upper bound to the amount of mineral-associated organic C that can be stored in soils (C_{sx}). This storage occurs through bonding of organic compounds to polar mineral surfaces, and through inclusion of particulate organic C during microaggregate and macroaggregate formation and turnover (Kong et al., 2005). The parameterization of the storage capacity and its influence on C_s dynamic is empiric, but the conceptual framework resembles the Langmuir adsorption isotherm or Michaelis-Menten dynamics. Organic soil N pools mirror the C pools. The C:N ratio of each pool varies and depends on the properties of the substrates being decomposed by microbes and the availability of mineral N. The C_s and N_s are only of microbial origin (Mazzilli et al., 2014). Particulate organic matter dynamics is lumped with that of plant and manure residue pools, which decay with first-order kinetics. These pools are replenished by crop residues and manure inputs resulting from crop growth and management practices implemented in any given simulation (e.g., residue removal, manure additions). The soil temperature, moisture, and aeration control turnover rates. These rates are upregulated by the degree of soil disaggregation by tillage using empiric equations that represent aggregate breakup and exposure of C_s to microbial degradation, and down-regulated by aggregate formation that returns turnover rates to a baseline (Kemanian and Stöckle, 2010; Pravia et al., 2019).

Empiric data available to support a saturation-dependent turnover rate is scarce, but the C_s dynamic is correctly simulated with this formulation (Kemanian and Stöckle, 2010; Pravia et al., 2019). The soil C stabilization fraction after microbial degradation (also known as humification fraction) also depends on the saturation ratio C_s/C_{sx} . The formulation that represents microbial biomass is described explicitly in White et al (2014) and with minor modifications implemented in Cycles. Summaries of this formulation were presented in Kemanian et al (2005), Kemanian et al (2011) and Pravia et al (2019), including comparisons with soil carbon measurements. A major advantage of the Cycles C_s cycling formulation seems to be the correct simulation of the C_s distribution with depth (e.g., Pravia et al., 2019), without resorting to depth-forcing functions that bypass mechanisms (e.g., Oleson et al., 2013). Steady state formulations of saturating behavior can be found in Kemanian et al (2011) and White et al (2014). For completeness, the equations to calculate the soil organic C and N balance are presented in SI Section V.

Nitrogen both mirrors and controls the C_s turnover rate and C:N ratio. The decomposition rates of N pools are coupled to the decomposition of the corresponding C pool (White et al., 2014). The C:N ratio of the microbial biomass depends on the availability of mineral N and the C:N ratio of the decomposing substrate (SI Section V). Decomposed N that is not assimilated by microbial biomass is transferred to the ammonium N pool (net N mineralization). Ammonia oxidation by comammox bacteria is not simulated explicitly. When the decomposing C pools do not contain sufficient N to meet the microbial demand, the

microbial pool assimilates N from the ammonium and nitrate pools instead (net N immobilization). If there is insufficient inorganic N to meet the microbial demand, the decomposition of organic materials is halted.

2.7. Agroecosystem features of Cycles

In agroecosystems, crop rotations can be a sequence of non-overlapping monocultures, or combinations of polycultures planted and harvested simultaneously (haying) or asynchronously. Among them are rotation of annual crops with multi-species pastures, cover crops planted into living monocultures (interseeding), intercropping, perennial pastures planted under nursing hay or grain crops, flushes of weed emergence and other multi-species canopies. Any of these combinations can be simulated with Cycles. Competition for light is based on assuming a randomly distributed canopy of leaves. The transpiration demand for each component of the community depends on the radiation interception, and the realized transpiration depends as well on the species rooting depth and root distribution. The water uptake (transpiration plus water redistributed through hydraulic lifting) of each species is estimated simultaneously assuming that all species operate at the water potential of each soil layer at the beginning of the time step. The water flux from the soil in each layer to the corresponding species also determines the access of each species to mineral N. The interaction between the distribution of energy in the canopy, which determines growth but also the potential water stress, and access to water and soil nutrients, leads to competition as an emergent property that, for example, determines the balance between legumes and grasses in a pasture (Pravia et al., 2019).

Field operations can be community- or species-specific, allowing for complex operation sequences. For example, a nursing annual grain crop that shades an underseeded perennial pasture stand, can be harvested for grain leaving a mixture of straw and an established perennial stand (e.g., Pravia et al., 2019). Tillage has multiple effects on soil and residues. In Cycles, tillage mixes all soil layer attributes based on a soil mixing efficiency coefficient specified for each tillage implement. In the long term, tillage tends to homogenize soil texture in the tilled layers. Lack of tillage causes stratification of C_s (and other properties), making C_s saturation simulation critical. A simplified summary of the simulation of tillage effects is available in Kemanian and Stöckle (2010). Surface residues alter the allocation of latent heat to the soil surface. Snow affects soil heat flux. Snow cover reduces cold damage to leaves depending on crop specific parameters.

2.8. Code structure, data input and data assimilation at runtime

Cycles is written in C. In the code, each major module is encapsulated within its own function. Only the state variables and parameters relevant to each module are passed along. Sub-modules within each function further organize the processes. This modular approach ensures reusability, ease of modification, and the flexibility to remove or replace modules in future developments. It also allows updates to be made to specific parts of the model without affecting other modules. It uses C structures to represent elements that require multiple instances such as crops growing concurrently. Inputs are kept to a bare minimum.

Weather inputs were described in the energy balance section. Soil data require information by horizon or user-defined layer. The information per layer includes texture, bulk density, rock content, initial C_s and N_s , mineral N, and other properties. Hydraulic properties can be user-defined or estimated with pedotransfer functions (Saxton and Rawls, 2006). Soil property initial conditions can be estimated by running simulations to a dynamic steady state when soil properties oscillate around a mean (no soil erosion or soil genesis). Default parameters are provided for dozens of crops. Field operations including tillage, planting, fertilization, harvesting, and irrigation are inputted in an operation file submitted at run-time. These operations can be set to a

fixed date or be conditional to soil temperature, soil moisture, and crop phenology thresholds.

Any model variable can be updated at run-time using a reinitialization file. This is useful, for example, to soft-couple models or force external information (e.g., Khan et al., 2019; Shi et al., 2023). A hydrology model may provide the water table depth in a landscape segment, which can be forced into Cycles. The soft coupling breaks down the balance of energy and mass at runtime and must be used carefully. However, these approaches also represent the conditions in which models like Cycles can be used in decision support systems (Zhai et al., 2020; Gil et al., 2021).

Cycles is a computationally efficient model that can be run on personal computers. However, for optimal performance, high-performance computing (HPC) systems can be utilized. For instance, the 3-D version, Cycles-L, leverages OpenMP (Dagum and Menon, 1998) for multi-thread processing, significantly enhancing computational efficiency (Shi et al., 2023). Additionally, shell and Python scripts can automate the generation of input files for large-scale applications; job management systems, such as Slurm (Yoo et al. 2003), allow users to submit multiple Cycles simulations across different processors (Ferreira Da Silva et al., 2019), enabling simultaneous execution and accelerating the overall process.

2.9. Test cases

Test cases show the skill with which Cycles simulates key process that have not been highlighted in prior work: 1) plant density effect on radiation interception of barley (*Hordeum vulgare* L.); 2) ET of maize (*Zea mays* L.) and shrub willow (*Salix* spp.); 3) net carbon ecosystem exchange (NEE) and daily CO₂ acquisition by perennial canopies; 4) winter cereal growth, grain yield, and N uptake; and 5) soil moisture depletion by soybean (*Glycine max* [L] Merr.). A brief description of each dataset follows with more information provided in the SI section.

3. Model evaluation

3.1. Modeling plant density and radiation interception of barley

Managing plant density is often relevant in water-limited environments. A comparison of the daily radiation interception of spring barley at two planting densities in a Mediterranean-type environment in Eastern Washington illustrate the application of Eq. (7). Details of the experiment are reported in Kemanian et al. (2004), and here we present

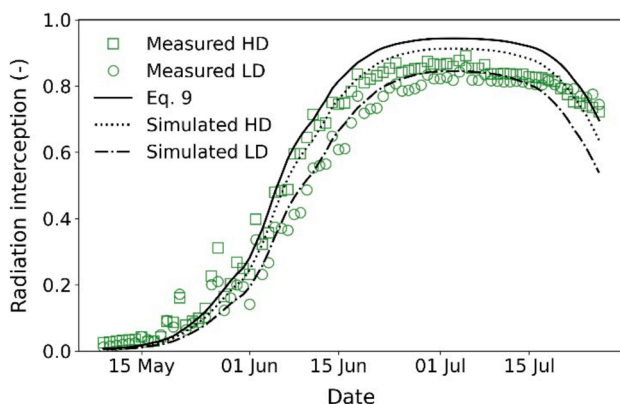


Fig. 3. Measured and modeled daily fractional interception of solar radiation for barley in 2000, at the Palouse Conservation Farm near Albion, WA. Both the high density (HD) and low density (LD) (180 and 105 pl/m²) are below the assumed reference optimum plant density (250 pl/m²) represented in Eq. (7). The maximum canopy cover (ϵ_{ix}) is 0.95. Experimental details in Kemanian et al (2004).

data from one cultivar and one planting date at two densities. The cultivar was Steptoe, a six-row genotype with upright habit and short cycle. The densities were 105 and 180 plants m⁻². The experiment had four replications. Radiation interception was measured with 0.5-m solarimeters installed in each plot and from measured incoming radiation. For the computations, it was assumed that a density of 250 pl m⁻² represents an optimum density for spring cereals, which overall show a low yield response to this variable (Fischer et al., 2019). Weather data to compute thermal time and cumulative solar radiation were downloaded from the North American Land Data Assimilation System (NLDAS; Xia et al., 2012).

Cycles simulations reproduced well the observed radiation interception of barley. The measured fractional daily radiation interception (ϵ_i) is presented in Fig. 3 as the average for each treatment. The cumulative radiation interception estimated using measured ϵ_i and S_i for the high- and low-density treatments were 1154 and 1065 MJ m⁻², respectively, and those using Cycles were 1165 and 1040 MJ m⁻². On a daily basis, the ϵ_i relative root mean squared error was 9 and 11 %, and the average bias was only 0.02 and -0.013 for high- and low-density, respectively.

3.2. Measuring and modeling ET of maize and shrub willow

While maize is a C₄ annual crop that requires no introduction, shrub willow is a lesser-known C₃ crop. Interest in willow as a biomass crop relates to its adaptation to climates with cold winters and temperate summers like those of the northeastern United States (Montes et al., 2021). Among willow's distinctive features is that it is a perennial for which budbreak occurs early in the season and on branches that are already developed, greening up rapidly a month before maize planting. We measured ET and NEE of maize and willow in two adjacent fields of 15-ha each at Rockview, Pennsylvania, using both open and closed path eddy covariance systems. Details of the experiment can be found in Cangiano (2015), Montes et al (2021), and in SI section VI. We compared the measured and simulated ET for 2014, a year in which the willow canopy was healthy, fully established, and in which timely precipitation enabled growth without water limitations for both crops.

Measurements show that the ET of shrub willow early and late in the growing season was higher than that of maize (Fig. 4). In this region, willow budbreak precedes maize planting by 4 weeks, and maize leaf senescence precedes willow leaf fall by 4 weeks in autumn. Cycles captured the lower ET of maize compared with willow in both the beginning (May) and end of the growing season (October). Cycles also captured the comparable ET of both crops at full canopy development and the ET temporal patterns (Fig. 4, Fig. SI.3). The modeled ET was higher than that measured, mostly when there were no green leaves early and late in the season (Table 2). The modeled ET was also slightly higher than the measured one from July to September, when the canopies fully covered the ground (Table 2). In that period, the difference between measured and modeled ET averaged 0.38 mm d⁻¹ (about 1 MJ m⁻² d⁻¹) (Table 2). Nonetheless, for both measured and modeled ET, the difference in cumulative ET between crops was similar (145 mm d⁻¹ for measured vs 155 mm d⁻¹ for modeled ET). These differences result from measured (modeled) ET from April to December of 500 (667) and 658 (813) mm d⁻¹ for maize and willow, respectively.

3.3. Modeling daily carbon acquisition of shrub willow

Comparing measured and modeled daily C acquisition is difficult because of the contribution of heterotrophic respiration to the CO₂ flux measured with eddy covariance techniques. In the same experiment in which we measured maize and willow ET, we measured CO₂ exchange between the atmosphere and the plant-soil system. The contribution of respiration was substantial in the maize field (Cangiano, 2015) due to the recent conversion of that field from a hay field to annual cropland. However, a comparison of modeled and measured C exchange was

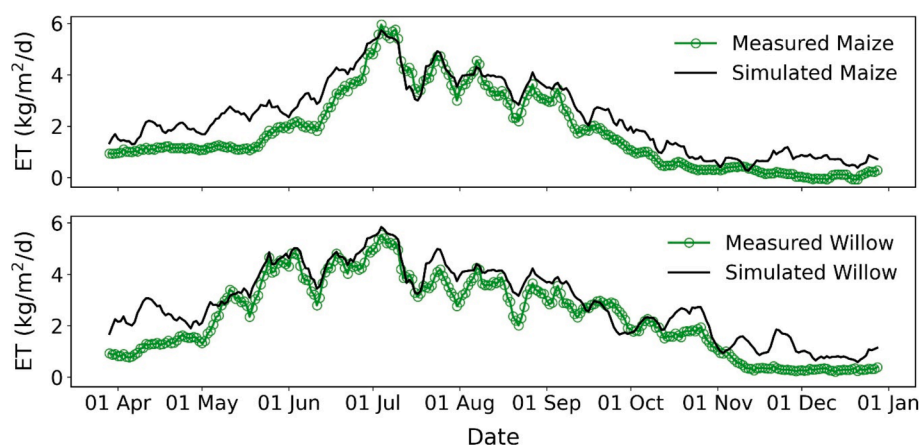


Fig. 4. Measured and Cycles-modeled evapotranspiration (ET) for maize and willow in adjacent fields at Rockview in central Pennsylvania, in 2014. Data presented as centered 7-day moving averages.

Table 2

Monthly averaged measured and Cycles-modeled ET in adjacent fields of maize and willow at Rockview, Pennsylvania, in 2014. Bold rows represent the period with fully developed canopies.

Month	Maize			Willow		
	Measured	Modeled	Mod. – Obs.	Measured	Modeled	Mod. – Obs.
mm d⁻¹						
Apr	1.11	1.79	0.68	1.22	2.35	1.13
May	1.34	2.52	1.17	3.12	3.56	0.44
Jun	2.85	3.69	0.84	4.19	4.57	0.39
Jul	4.53	4.48	-0.05	4.16	4.61	0.45
Aug	3.30	3.74	0.44	3.30	3.93	0.63
Sep	2.11	2.79	0.68	2.81	2.86	0.05
Oct	0.58	1.15	0.58	1.73	2.26	0.54
Nov	0.27	0.71	0.44	0.48	1.27	0.79
Dec	0.06	0.68	0.62	0.31	0.85	0.54

possible for the willow field that was established in 2012 because it had low heterotrophic respiration before budbreak (winter) and at night-time, showing net CO₂ uptake most of the growing season.

The measured NEE and the Cycles estimated CO₂ gain show reasonable agreement (Fig. 5), indicating that Cycles simulates CO₂ acquisition well. The model might be underestimating canopy cover and growth in May, early in the growing season, as the simulated CO₂ acquisition seem to lag the measured one (we did not “calibrate” to reduce this deviation). The modeled high and low CO₂ uptake peaks are subdued compared with measurements. A strict evaluation of NEE would require estimating and adding heterotrophic respiration to NEE, but as indicated in the experiment description, heterotrophic respiration in this established shrub willow field is low.

3.4. Modeling growth, yield, and nitrogen uptake of winter and spring cereals

Crop and soil display pronounced spatial variability (e.g., Huggins et al., 2014). The challenge of representing landscapes in 3D has been already alluded to, but nonetheless, it is useful to test to what extent a 1D model can represent field averages when the variability is known. We used Cycles to simulate the yield of cereals in a large experimental field with complex terrain, the 34-ha Cook Agronomy Farm of Washington State University. This field is in a rainfed, semi-arid location, with an annual average precipitation of 550 mm with a Mediterranean seasonal pattern. Soils are deep silt loams developed on loess deposits, but topography and hillside aspect generate a gradation of soil moisture and radiation environments that coupled with soil variations causing substantial within-field yield variability. The field was planted with spring wheat (*Triticum aestivum* L.) in 1999 (cv. WB926R.), spring barley in 2000 (cv. Baroness), spring wheat in 2001 (cv. Hank), and winter wheat in 2001 (harvest 2002, cv. CDC Falcon). In 1999 and 2000, wheat and barley were planted in the entire field, and in 2001/02 the field was divided into strips and data from the 12-ha central strip of spring wheat and winter wheat are presented here. Biomass and yield were sampled on a fixed grid in 1999 and 2000 (ca. 4 sampling points per ha), and along a transect in the central strip of the field in 2001 and 2000. In addition, and for winter wheat only, sequential biomass samples were taken during the growing season at four preset locations across a N-S transect that captured from a north facing backslope to a footslope on the south side of the hill. Since Cycles is a 1D model, simulations used a Palouse silt loam typical for the region. Details provided in Huggins et al. (2014) and SI section VII.

For the four cereals, Cycles simulation results compare well with the average harvested biomass, grain, and straw (mostly within 10 % of the

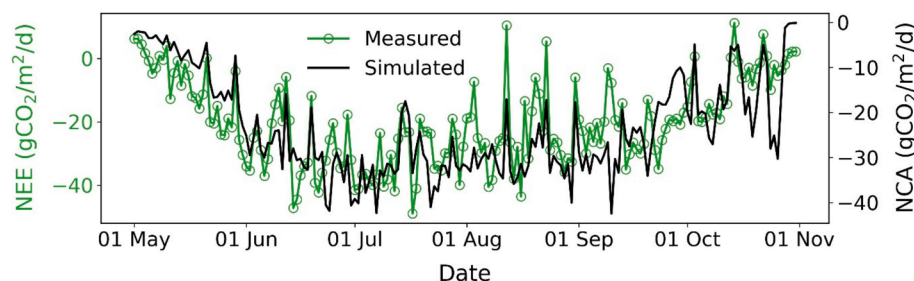


Fig. 5. Daily measured net ecosystem carbon exchange (NEE) and Cycles-modeled daily net carbon acquisition (NCA) in equivalent units for willow at Rockview, Pennsylvania, in 2014.

Table 3

Measured and modeled aboveground biomass, grain, straw, nitrogen (N) mass in each fraction, and the biomass harvest index (HI) and nitrogen harvest index (NIH). The crops were grown sequentially in the Cook Agronomy Farm, WA, in 1999, 2000, 2001, and 2002, and modeled in the same way with Cycles. Italicized numbers are the standard deviation of the observations (WB926R n = 140; Baroness n = 140; Hank n = 26; CDC Falcon n = 16).

	Spring Wheatcv. WB926R		Spring Barleycv. Baroness		Spring Wheatcv. Hank		Winter Wheatcv. CDC Falcon	
	Meas.	Model	Meas.	Model	Meas.	Model	Meas.	Model
Biomass	kg/ha							
	6430	6630	10,730	9600	9500	9640	11,910	11,370
Grain	1250		1700		1120		2060	
	2590	3170	4570	4270	4440	4310	4950	4580
Straw	560		740		520		950	
	3840	3460	6150	5330	5090	5330	6960	6790
N biomass	760		1090		800		1210	
	78	103	103	172	108	136	138	110
N grain	18		24		15		19	
	62	71	75	114	90	92	108	73
N straw	15		18		13		7	
	15	32	28	58	18	44	31	47
HI	5		9		5		23	
	kg/kg							
NHI	0.40	0.48	0.43	0.44	0.47	0.46	0.41	0.40
	0.03		0.03		0.03		0.03	
	0.80	0.69	0.73	0.66	0.83	0.68	0.71	0.66
	0.04		0.05		0.03		0.04	

average of the observations), and with N in grain (Table 3). However, modeled results tended to overestimate N in straw and in total biomass, and underestimate translocation of N to the grain, as the measured N harvest index ranged from 0.71 to 0.83, and the simulated one ranged from 0.66 to 0.69 (Table 3).

The growth and N uptake dynamics are shown in Fig. 6 for winter wheat only. The model represented the growth dynamic of winter wheat well, on average, but the simulation set up requires comments. Crop canopy cover is critical to simulate energy and mass exchange with fidelity as it affects the proportion of the atmospheric evaporative demand that could be dissipated via canopy transpiration. Since we had measurements of ϵ_b , the plant density in this simulation was adjusted from the default $PD_f = 1$ to 0.5, to match observations. The reason is that this hard red winter wheat cultivar grows slowly early in the season, is extremely upright thus delaying soil cover, and in local variety trials yielded only 85 % of the average trial yield (WSU Extension, 2002).

The variation in biomass and other variables within each sampling date (Fig. 6) reflects water redistribution due to topography and the corresponding plant response. In 2002, precipitation events after May only moistened the soil top 0.1-m, and thus winter wheat relied on stored soil water. As a result, mineral N applied in spring remained in the dry topsoil and out of reach of the active plant roots (SI Figure SI.5).

Lower landscape positions that remained moister through the growing season produced more biomass and recovered more N not only from fertilizer, but from N mineralization. Dry years allow water drainage and oxygenation of low landscape positions that often have a deep A horizon due to deposition of upland erosion.

Together with earlier publications that show agreement between measured and modeled grain and forage yield (Pravia et al., 2019), and that the approach to compute growth used in Cycles has been used (with minor modifications) in other models (Stöckle et al., 2014; Castaño et al., 2021), these results indicate that Cycles simulates plant growth well.

3.5. Modeling soil water depletion under soybean

When modeling soil water depletion, there are critical factors related to soil–plant interactions that affect model performance but are independent of the water uptake algorithm. Among them are the maximum rooting depth, the speed of the root penetration front, and soil attributes that define field capacity, wilting point, and water redistribution, and plant and soil attributes that affect root exploration like macropore network and rock fragment distribution, among others. Dry years are most useful for model testing as they help define effective rooting depths

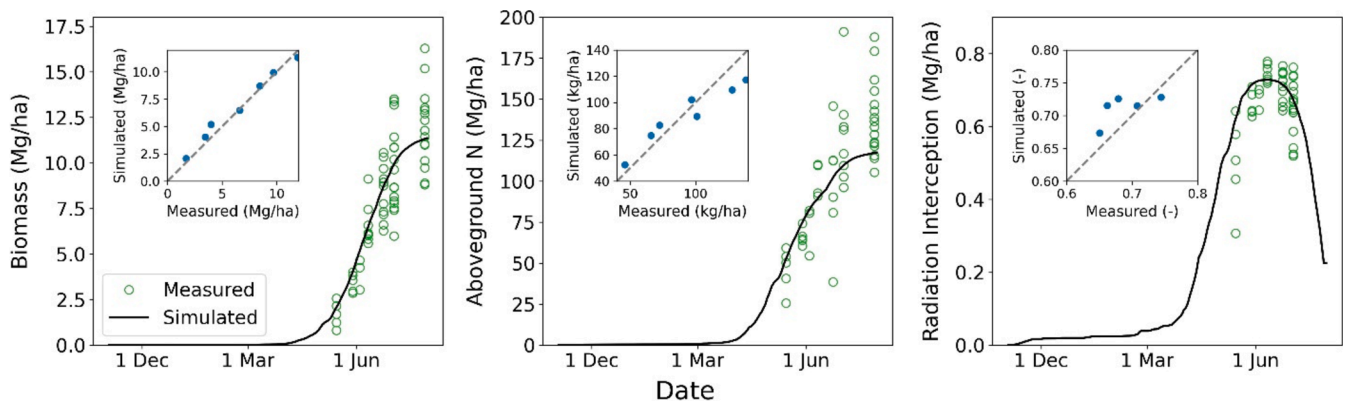


Fig. 6. Measured and modeled aboveground biomass (top), N(middle), and solar radiation interception (bottom), for winter wheat. Cultivar CDC Falcon, planting year 2001, location Cook Agronomy Farm, WA. The N fertilization was split 80/60 kg/ha of N in fall at planting and in spring. Insets show average modeled vs average measured values in a 1:1 frame; the averaging controls for the increase in variance as crop growth progressed. The slope, intercept, and r^2 of the modeled vs measured variables were 0.89, 0.97 kg/ha, and 0.99 for biomass (n = 7), 0.64, 30 kg/ha, and 0.92 for aboveground N (n = 7), and 0.47, 0.40, and 0.87 for radiation interception (n = 5), respectively.

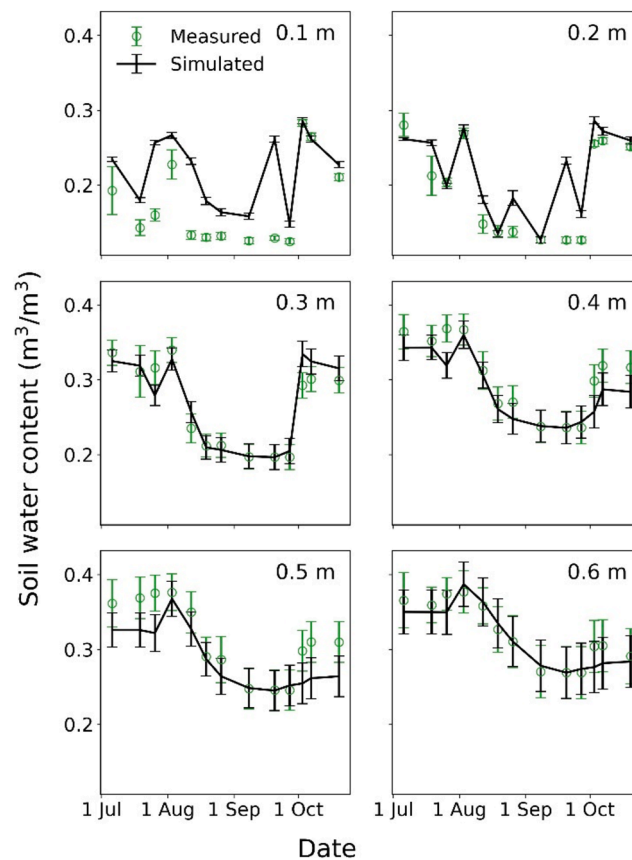


Fig. 7. Measurement and simulation of soil moisture in Landisville, PA. Soil moisture was measured with a Diviner and modeled with Cycles. Data reported from the week after emergence to harvest. Soybean was planted on 26 June 2016, following winter wheat. The model was run for each replication and results averaged. Error bars indicate the standard deviations of five replications.

and water depletion profiles. In 2016, we collected such information in an experiment with double-cropped soybean in southeastern Pennsylvania, at the Penn State Southeast Agricultural Research and Extension Center in Landisville, a region with some of the highest soybean yields in the USA (Stefani-Faé et al., 2020). The experiment had four replications and soybean followed winter wheat. The region average temperature is 9.5° C and precipitation is 1040 mm y⁻¹. Soybeans were planted after harvesting winter wheat on 26 June 2016. Soil water content was measured weekly from planting to maturity using a capacitance probe (Diviner 2000, Sentek Pty Ltd, Stepney, Australia) that reported data at 0.1-m soil depth intervals to a depth of 0.6 m. The sensor readings were converted to soil water and compared with those of Cycles. Soybean growth and yield was simulated for each block independently to account for differences in soil texture and initial wheat residue load. Other details can be found in SI Section VIII.

Soybean soil moisture depletion and recharge in southeastern PA was modeled accurately with Cycles. The main disagreement between observed and modeled soil moisture occurred in the topsoil (0.1 m), where measuring this variable accurately is challenging and observations should be interpreted with caution. The higher responsiveness of the top layers to precipitation indicates that preferential flow through macropores was limited. The model represented very well the sequential depletion of water from the 0.3 to 0.6 m layers, a feature that can only be observed in the field when soils are at field capacity at planting and have limited recharge during the growing season. Stefani-Faé (2019) also reported excellent agreement between measured and modeled soybean yield for a dataset that included this location.

3.6. Application: An autonomous crop sequence builder

The application presented here highlights the ability of a Cycles-based system to execute thousands of simulations, represent distinct rotations, and develop tools relevant for system analysis and design. We introduce Cycles-A, an autonomous crop sequence builder, along with an application to develop crop sequences in the cropland of the Chesapeake Bay Watershed. A second application, presented in SI Section IX, illustrates the use of Cycles to develop a twin-model of the > 870 k fields cropped in the state of Iowa and displays the results for surface runoff.

Cycles-A requires a domain definition consisting of a geolocated field that is matched with a weather file, a soil file, a portfolio of crops that can be planted at that location, and their grain price and production cost. Thereafter, Cycle-A works as follows: 1) Generate matrices of crop yields and growing season lengths for all plausible planting dates in all simulation years for each crop selected in the crop portfolio. These matrices are created by looping through planting dates and running each crop across all simulation years using Cycles. 2) Start crop sequence generation. If no crop planting event is in the calendar (time 0), Cycles-A evaluates all combinations of two crops planted back-to-back (C1 and C2) on all planting dates using the matrix of crop yield and growing season length. The yield and growing season-length of each crop on a specific planting date is sampled randomly from the simulated matrix for each plating date, thus introducing stochasticity. For each combination of crops C1 and C2, Cycles-A calculates the daily economic return for the crop sequence, accounting for crop yield, growing season length, crop prices and production costs. The two-crop combination with the highest daily economic return is selected for planting, and crop C1 assigned as the next crop to be planted. 3) Once crop C1 is selected, the tillage and fertilization operations associated with that crop are automatically

added to the schedule of field operations. 4) After planting and harvesting (or killing) crop C1, the planting of the next crop is re-evaluated by re-activating the autonomous crop sequence builder, as the planting of crop C2 is no longer guaranteed to be the best option. This step accommodates dynamic changes in prices, and accounts for the effect of the simulated harvest date of C1 in defining future options. Potential restrictions can be specified, such as a minimum or maximum frequency of a crop in the overall crop sequence to meet farm management logistic, pest and weed control needs, or regulatory compliance. Cycles-A can be expanded to include functions to weight crop choices using an opportunity cost to account for longer crop sequence that includes multi-year pastures (Ernst et al., 2016), and irrigation and fertilization needs.

We used Cycles-A to explore the most suitable crop sequences across the Chesapeake Bay Watershed. Within the cropland area of each county, only one representative soil and field per county was selected for these simulations. The crop menu had winter wheat and three maize and soybean maturity types, totaling seven crops. The maize relative maturity (RM) types were RM90, RM100 and RM110, and the soybean maturity groups were 3, 4 and 5. Maize for silage and alfalfa were not considered in this example, as they are subordinated to dairies and their forage demands, but these crops can be included in a more comprehensive analysis. The only restrictions imposed were that soybean could not follow soybean and that planting was conditional to the temperature and soil-moisture in the top 10 cm. Results are presented as maps for the Chesapeake Bay Watershed.

Cycles-A, the autonomous crop sequence builder based on Cycles, produced crop sequences that resemble those observed in the Chesapeake Bay Watershed, with double cropping in the south and single cropping in the north of the watershed (Fig. 8). When double cropping, shorter season maize and soybean increased in frequency northward to accommodate winter wheat. Maize RM110 was only selected in the southern portion of the watershed, while maize RM95 replaced longer season maize in the northern portion of the watershed. The timeline in each region (SI section X, Fig.SI9) shows an increase in fallow time northward, suggesting time windows where cover crops could be accommodated. It is simple to add a monetary value to a cover crop (a green payment) and integrate it in Cycles-A to explore which incentive would stimulate cover crop expansion and in which region.

4. Discussion

More than 40 years ago, Monteith (1978) stated that “complexity [in models] is rarely achieved without recourse to chains of assumptions”

and added in a subsequent paragraph that “the tendency for models to become more complex should be balanced by attempts to identify and eliminate inputs or relationships that turn out to have little bearing on the outputs”. In developing Cycles, we have followed this credo. As illustrated in the model description and test cases, we focused on generalizations and avoided the allure of adding detail when it comes with high parameterization costs or unrealistic calibration demands. We developed a code structure that allows both expansion into a 3D landscape model and data assimilation at runtime. The result is that Cycles is accurate, easy to use and develop further, and non-ag experts can integrate it into complex workflows (Ferreira da Silva et al., 2019; Gil et al., 2021; Turchetta et al., 2022), run it in ca. 1 million fields (SI Section IX) or embed it in optimization algorithms (Fig. 8).

Nonetheless, operating agroecosystem models competently still requires both training and disciplinary expertise (Confalonieri et al., 2016), because input data quality varies, the concept of weather forcing can be too rigid, model parameters are correlated, and relevant interactions might not be accounted for in the model algorithms. These limitations are not always clear for model users and model developers alike but can be bypassed with regression or machine learning modeling. The impressive success at predicting yield within the domain of the training dataset of regression model (Schlenker and Roberts, 2009) or inductive machine learning methods (Hoffman et al., 2020), resides in the fact that these methods bypass some of these impedances. In addition, machine learning models can use input information more fully than mechanistic models, developing their own transfer functions instead of relying on the transfer functions embedded in mechanistic models (e.g., Liu et al., 2022).

The concept of weather forcing, as used in Cycles and all agroecosystem model, can be inadequate if simulated conditions depart from surface conditions implied by the weather forcing (e.g., simulated irrigation in an otherwise rainfed environment). In that case, important deviations between observed and simulated energy fluxes can occur due to lack of land-atmosphere feedback.

When modeling the effect of plant density on radiation interception, we show that a generic approach in Cycles (Eq. (7)), accounts for most of the differences in radiation interception without parameterizing, for example, changes in barley tillering which is species- and cultivar-specific. The simulations were not perfect. While the bias was near zero, the daily relative mean square error was close to 10 %. Among the reasons for such discrepancy is that we did not account for the fraction of diffuse and beam radiation due to hourly changes in radiation load or cloudiness. There is, however, no assurance that a more detailed

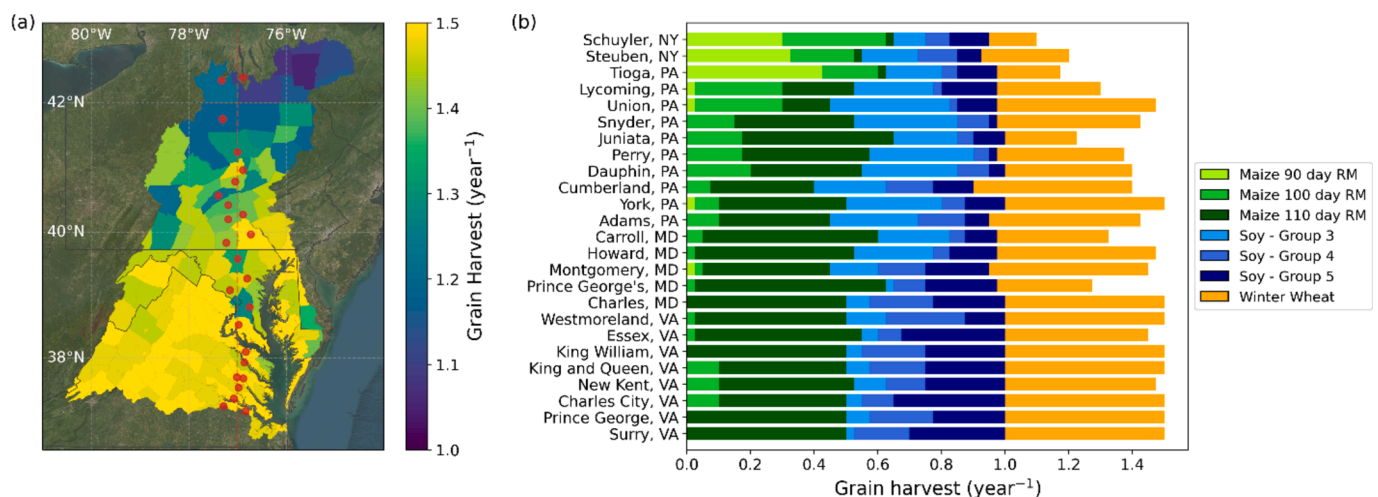


Fig. 8. Realized crop sequences obtained across by running Cycles-A, an autonomous crop sequence builder, from 1980 to 2015 in all counties of the Chesapeake Bay Watershed. Seven crop options were allowed, maize (RM90, RM100 and RM110), soybean (maturity groups (3), 4 and 5), and winter wheat. (a) Grain harvests per year; red dots mark counties along the 77° meridian. (b) Frequency of each crop in counties along the 77° meridian.

radiation interception model will improve the simulation results. A major advantage of the approach used in Cycles is that it makes simulating mixed stands feasible without depending on a complex parameterization of canopy architecture.

Another challenge when using agroecosystem models is accounting for the potential correlation among parameters. For example, simulations of transpiration that rely on the parameterization provided by the Penman-Monteith equation for daily water demand calculations contain two limitations that are difficult to avoid and can only be partially ameliorated by other methods. First, while the solution to the Penman-Monteith equation respects energy conservation, the multiplication by the crop coefficient (k_c) rearranges energy allocation without much compunction. Second, using weather inputs as forcing assumes no feedback to the atmosphere. The first limitation can be addressed by simultaneously solving the vapor surface conductance (g_{sc}), boundary layer conductance (g_{bl}), and energy fluxes, but without solving the second limitation. In addition, any approach that improves upon the Penman-Monteith equation with or without solving for g_{sc} , needs to account for the effect of transpiration on g_{sc} (Monteith, 1995). Yet, the approach is still useful, as Christy et al (2018) represented well the hydraulic behavior of two contrasting wheat cultivars, even though in their model there is no transpiration-conductance feedback. The typical g_{bl} in low to medium wind conditions is between 0.5 to 2.5 mol m⁻² s⁻², while that of highly transpiring crops can be as high as 1.3 mol m⁻² s⁻² (Kimm et al., 2020) or slightly higher at $D_a < 1$ kPa. The importance of this observation is that the operative g_{sc} can depend on g_{bl} , as increasing g_{bl} may force stomatal closure (Monteith, 1995; Lhomme et al., 1998). These controls on transpiration (and therefore growth) become more relevant in conditions of high D_a or high wind speed.

Interestingly, this effect is partially accounted for in Cycles by using Campbell's (1985) approach to model transpiration. In Campbell's approach, when a preset maximum liquid water flux is exceeded, stomata start closing thus limiting transpiration. Some limitations remain, as in practice, the resistance to water flow for a given genotype is not constant through the life cycle as xylem vessel diameter and other resistances vary and can change with the environment. The comparison of daily measured and simulated ET using Cycles indicates that daily time step models can accurately simulate ET (Fig. 4), and agrees with similar comparisons (Khan et al., 2019).

The radiation- and transpiration-based approach used in Cycles for modeling CO₂ acquisition is robust as it limits growth depending on light availability or water availability, even though G_R and G_T are correlated as radiation and temperature are embedded in the Penman Monteith equation (but without an adjustment of g_{sc} by temperature, an adjustment that we included *ad hoc*). Using a coupled photosynthesis and transpiration model (Collatz et al., 1991; Bassiouni and Vico, 2021) would bypass some of these limitations. But again, it would also transfer uncertainties to an unwieldy number of parameters that are difficult to obtain for many species or cultivars. As currently formulated, a limited number of parameters in Cycles, like in Christy et al (2018) or Khan et al (2019), allows representing contrasting species or cultivars with simplicity and sufficient detail for agroecosystem-level analysis.

Modeling water use correctly both allows and depends on modeling soil water storage correctly. In Cycles, modeling water infiltration rate as the byproduct of runoff suffices to provide correct simulations of soil moisture (e.g., Fig. 7). The limitations of the CN method, originally developed to simulate runoff from watershed, have been extensively discussed in the literature (Ogden et al., 2017). However, the discussion of the method limitations in the literature has not been balanced by a discussion of the limitations of other methods that contain simplifications, assumptions, and operational challenges.

The simulation of water infiltration and redistribution based on first principles is challenged by variations of soil properties that are unknown or poorly described, despite ongoing progress in 1D or 2D models (Wang et al., 2020; Wang et al., 2021). Surface properties and soil porosity change with tillage (Lipiec et al 2006). Macropores develop gradually in

no-till soils, sometimes with accompanying but inconsistent changes in bulk density (Kay and VandenBygaart, 2002). Plow pans can develop in tilled soils (Unger and Kaspar, 1994), and a "platy" soil structure can develop near the soil surface under no-till due to traffic and surface compaction (Sasal et al., 2017). These and other alterations of the soil affect hydraulic properties in ways unaccounted for in pedotransfer functions (Cheng et al., 2018; Stefani-Fae et al., 2020). Thus, methods for numerically solving non-linear formulations of the water flow that come with large computational and parameterization costs often fail to converge to a solution (Campbell, 1985; Romano et al., 1998; Romano, 2014; Van Dam and Feddes, 2000). More importantly, they provide mathematical comfort without resolving the uncertainty in modeled fluxes that depend on hydraulic properties. Even if numerical convergence were achieved, water flow is often dominated by macropore flow, tile drain flow (a form of macropore flow), and lateral movement of water perched atop impermeable or saturated layers, all factors for which global parameterizations are not available. Amenable and practical simplifications are needed in agroecosystem modeling, while recognizing that 1D models and our knowledge base are limited (e.g., Timlin et al., 1996, 2002; Garen and Moore, 2005).

Some, but not all simplifications, can be removed when using a 3D domain, as infiltration, capillary rise, and exfiltration can be solved simultaneously in each land grid and river segment of the watershed (Shi et al., 2013). In such a case, however, the macropore structure including tile drainage needs to be calibrated. Macropore flow is critical in sloping terrain (Shi et al., 2013), and tends to be more relevant than minor unsaturated water transfers between soil layers. Using a numerical method that does not consider macropore flow may neglect the dominant control of infiltration. Furthermore, the effective k_{sat} may depend on biopore formation, collapse, and regeneration, all processes that as of today are not predictable. In lieu of this complexity, both Green and Ampt and the CN method provide a simple bridge to estimate infiltration in 1D models.

The soil C and N cycling in Cycles has been discussed by White et al (2014) and Pravia et al (2019), and we emphasize here that including the concept of C and N saturation confers Cycles a unique advantage and low parameterization demands. Improving the characterization of C saturation is of critical importance for a generalized understanding of organic C and N stabilization in a heterogeneous porous media like soil, as this feature is often neglected (Tao et al., 2023). Cycles future developments in this direction include a generalization of the nutrient cycling algorithm to include phosphorus (McConnell et al., 2020), sulfur, and other elements, and modeling erosion and deposition, which are more suitable for Cycles-L.

Cycles has limitations inherent to 1D models of SPAC, and limitations that are inherent to the assumptions in the model, some of which have been already alluded to. We recognize that using models with further detail in plant ecophysiology, soil physics, or plant-atmosphere coupling might be needed for other applications (Stöckle and Kemanian, 2020). The lack of a 3D domain limits the mechanistic modeling of the depth of the water table, and therefore if a water table is present within the rooting depth, it must be used as dynamic forcing that alters the water balance (potentially removing or adding water). For a single field, erosion and deposition cannot be simulated mechanistically: eroded material could be removed from the field, but no allowance is made for deposition. Without using a 3D model, deposition could be accommodated by allowing the input of soil fractions (a broadcast application of clay, silt, sand, stones, and organic matter), but not simulated. Air-filled porosity is used in Cycles as a surrogate for oxygen availability, but a simulation of oxygen consumption and diffusion is needed to simulate the dynamic of soil microbial communities or processes that respond to oxygen concentration. To that effect, the dynamic destruction and formation of macropores and changes in bulk density will need to be modeled dynamically. Many processes in row crops operate in 2D, for example, banding of N fertilizer. The banding of mineral N affects multiple processes, from crop N uptake to N leaching and nitrous oxide

production (Bowen et al., 2018). Within the framework of Cycles or any 1D model and considering field- or subfield-level applications (i.e., not watersheds), using a 2D grid to simulate soil and plant processes, as done by Timlin et al (1996) would be one of the most important steps towards improving the capabilities of Cycles; other efforts in this direction are mostly empiric (Hammer et al., 2009). Further progress in Cycles will consider expansion in the 2D domain and incorporating the simulation of biotic stresses and a more mechanistic consideration of the effect of the phytobiome on soil–plant processes.

The use of the model to build Cycles-A illustrates the power of the model for integration into data-model workflows. Cycles A can incorporate a richer menu of crops (or cycle durations), a time series for grain price and production cost for each crop, as well as regional forcings to account for the availability of services for planting, harvesting, grain transportation and other logistic constraints. In many regions, crop rotations are stable as they are adapted to the combined forcing of climate, markets, and other incentives. For example, in many states of the Midwest maize-soybean rotations dominate the landscape, and the increase in grain prices in the early 2020 s promoted an expansion of this rotation into grasslands (Wright and Wimberly, 2013), rather than changes in the rotation. But in other regions, particularly those transitioning between single and double-cropping, market swings or changes in the time window for operations can change the crop mix as the menu of crops is more varied. Furthermore, as climate change alters the time window for operations due to changes in precipitation patterns in this region (Wolfe et al., 2018), tools like Cycles-A can help explore crop sequences in fields and regions under a diverse set of forcings.

Applications that integrate Cycles into data-model workflows that allow runtime updates with real-time information will diversify the type of applications and continue advancing the premise of making models not just knowledge repositories or mathematical marionettes (Kirchner, 2006), but useful tools for landscape management.

5. Conclusions

Cycles is a repository of functional knowledge and a tool for agroecosystem analysis. The model simulates well radiation interception, biomass accrual, ET, soil water depletion, yield, and N uptake in both dry (Eastern Washington), and wet environments (Pennsylvania). It adds to existing literature showing Cycles' skill at simulating grain and forage yield as well as soil C and N dynamic. Among the distinctive features are its consideration of the current state of knowledge in soil biogeochemistry, ability to simulate any monoculture or polyculture sequence in combination with multiple management practices, and a forward looking objected-oriented structure that has been tested and integrated in data-model workflows. The model can seamlessly integrate into decision support tools, as demonstrated by the development of the autonomous crop sequence builder Cycles-A. The application domain for agroecosystem models is expanding. It is now possible to model every field in a watershed, region, or country, as well as subfields within fields, at time-scales relevant for management. The deployment of Cycles-L, a 3D version that simulates landscape processes, will further improve the representation of terrestrial processes at a scale relevant for interventions in agroecosystems. As the usage of 3D models unfolds, we expect that Cycles 1D will become a useful tool for educators, researchers, and practitioners interested in agroecosystems analysis.

CRedit authorship contribution statement

Armen R. Kemanian: Writing – original draft, Visualization, Supervision, Software, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Yuning Shi:** Writing – review & editing, Visualization, Software, Methodology, Formal analysis, Conceptualization. **Charles M. White:** Writing – review & editing, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Felipe Montes: Writing – review & editing, Visualization, Methodology, Formal analysis, Data curation. **Claudio O. Stöckle:** Writing – review & editing, Software, Methodology. **David R. Huggins:** Writing – review & editing, Software, Data curation, Conceptualization. **Maria Laura Cangiano:** Writing – review & editing, Visualization, Validation, Formal analysis, Data curation. **Giovani Stefani-Faé:** Writing – review & editing, Validation, Investigation, Formal analysis, Data curation. **Rachel K. Nydegger Rozum:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Credit

ARK conceptualized the model, developed the first version of Cycles in VBA, assembled eddy covariance data, collected data on winter wheat growth and yield, and prepared the first version of the manuscript. YS assembled the model in C, maintains the code in GitHub, operationalized the commands for multiple management operations and other user-friendly tools, and co-developed with ARK the plant density algorithm. CMW expanded the model to include soil microbial biomass explicitly, and conceived and developed the first version of Cycles-A, the autonomous crop sequence builder. FM and MLC installed and managed the eddy covariance equipment and analyzed and curated the raw data used in this paper. COS provided the original water uptake module for one plant species. DRH supervised data collection at the CAF and provided input in the development of the original carbon saturation model. GSF collected and analyzed the soil moisture data on soybeans, and prepared simulations and figures pertaining to soil moisture simulations. RKR developed the simulations, figures, and analysis for the application of Cycles to the cropland of Iowa. All coauthors edited the manuscript and contributed to data interpretation and discussion.

Access to the model

Download manual and executable: <https://psumodeling.github.io/Cycles/>

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.compag.2024.109510>.

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